

Carbon Offset Projects

What are carbon offset projects?

Carbon offset projects involve the verification of environmental conservation or protection activities that aim to diminish, prevent, or eliminate greenhouse gas (GHG) emissions, thereby contributing to the alleviation of climate change.

Carbon offset projects can be classified into two primary categories:

- 1. The initial category involves measures to prevent or decrease the release of carbon into the atmosphere.
- 2. The second classification centers on initiatives aimed at withdrawing carbon from the atmosphere by means of removal and sequestration.

Registrations and credits issuance on the Voluntary Carbon Market in 2022

Source: Climate Focus - 01/2024

Registered projects by activity type



Activity type	Value
Renewable Energy	36,55%
Household	35,35%
Nature-based Solutions	13,34%
Waste	8,95%
Industry	3,46%
Coal mine methane	1,24%
Energy (Non-renewable)	0,57%
Urban	0,51%
Gas recovery and utilization from oil fields	0,03%

Credits issued by activity type



Activity type	Value
Nature-based Solutions	40,04%
Renewable Energy	31,47%
Industry	12,28%
Household	8,86%
Waste	4,29%
Coal mine methane	1,73%
Energy (Non-renewable)	1,71%
Urban	1,33%
Gas recovery and utilization from oil fields	0%

Overview of nature-based solutions

Source: Climate Focus - 01/2024

Registered projects by activity type



Activity type	Value
Afforestation/Reforestation	48,63%
Improved forest management	22,26%
Avoided deforestation	18,75%
Avoided conversion	4,42%
Carbon sequestration in agriculture	2,9%
Reduced emissions in agriculture	1,83%
Watland restoration	1,22%

Credits issued by activity type

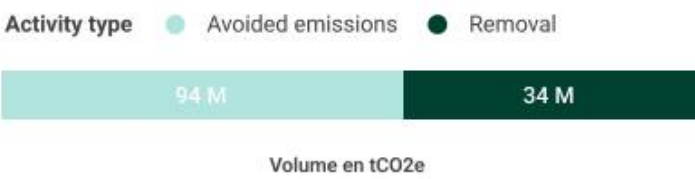


Activity type	Value
Avoided deforestation	49,33%
Avoided conversion	21,11%
Afforestation/Reforestation	15,72%
Improved forest management	6,1%
Carbon sequestration in agriculture	4,73%
Reduced emissions in agriculture	3%
Watland restoration	0%

Share of registered projects



Share of credits issued



One of the notable methods for carbon offsetting is through the adoption of renewable energy sources. This includes the installation of solar, wind, or hydro facilities globally. As of 2022, approximately 30% of the global electricity is derived from renewable sources like hydropower, solar, and wind, with a rapidly increasing demand for these sustainable energy options.

What qualifies a high-quality carbon project?

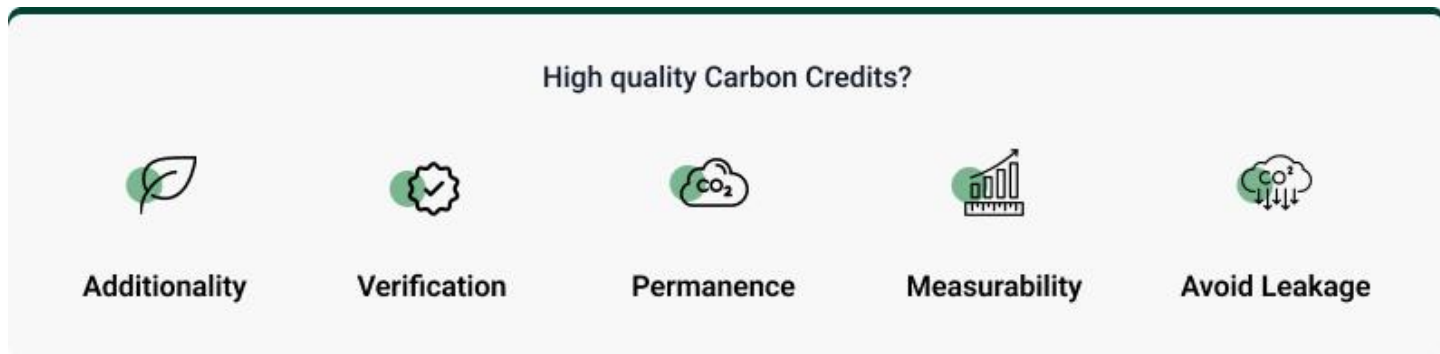
To meet the standards of a high-quality carbon offset project, several essential criteria must be satisfied. First and foremost is the principle of additionality indicating that the project's existence relies on the revenue generated from carbon credits.

Rigorous assessment is ensured through an impartial third party, responsible for continuous monitoring, reporting, and verification, a process integral to the project's credibility.

Permanence is another vital criterion, emphasizing that any achieved carbon reduction or removal should be irreversible.

Precision in measuring the project's impact is maintained through adherence to scientific standards and a recognized methodology, ensuring measurability.

Lastly, avoiding leakage is crucial, signifying that the project should not inadvertently lead to increased emissions elsewhere or account for any emissions that may occur. These criteria collectively uphold the integrity and effectiveness of high-quality carbon offset initiatives.



Who certifies Carbon Credits and how is it done?

Certifying carbon credits is a crucial process carried out exclusively by accredited organizations, and both compliance and voluntary carbon markets have distinct standards to uphold. These standards are enforced with the assistance of independent verification bodies, ensuring that carbon reduction projects adhere strictly to the rules governing offsetting.

A carbon offset credit achieves certification only by undergoing the specified processes or procedures outlined by the certifying standard. Commencing with project planning, similar to other projects, carbon offset developers must estimate the climate impact and assess it against the stringent standards set by the carbon credit certification body. Upon completing estimates and assessments, the next stage is project approval, entailing the initial review by the certification organization. If plans align with standards, the green light is granted to proceed.

Post-approval, the process advances to third-party project validation, where an independent assessment is conducted by the carbon credit certification company, such as Verra or GS. This comprehensive evaluation may include a site visit to validate compliance with certification standards. Upon successful validation, the project moves to the final review and approval stage, obtaining certification and being issued certified carbon credits. These credits can be sold through a carbon registry or used for offsetting emissions.

In continuation, the certified project requires project monitoring, involving the implementation of the monitoring plan developed during initial project planning. Regular monitoring project reports must be submitted to the carbon credit certification body, ensuring that the sold credits have effectively contributed to the reduction or removal of emissions. This meticulous process guarantees the integrity and ongoing impact of certified carbon offset projects.